Industrial Self-Collection App

Mobile & Web Platform for Recycled Material Collection Management

Proposal Date: November 06, 2025

Executive Summary

A comprehensive digital solution to automate the collection of recyclable materials from industrial partners. The app replaces manual coor

■ SIMPLIFIED PROCESS FLOW (5 STAGES)

Stage	Actor	Actions	System Response
1. Request	Vendor	• Creates vehicle request via app • Enters material type, quantity, date/time • Selects pickup location	• Generates request ID • Sends alerts to Transporter & POC
2. Approval	Transporter & POC	• Reviews request details • Checks vehicle availability • Assigns vehicle & driver • Approves/rejects/reschedules	• Confirms assignment • Notifies vendor of schedule • Activates GPS tracking
3. Collection	Vendor Transporter	• Vehicle arrives at vendor site • Material loaded onto vehicle • Weighing before & after loading • Vendor acknowledges completion	• Records gate entry • Captures weight data • Documents loading status • Updates request status
4. Transport	Transporter	• Vehicle tracked via GPS • Arrives at BSP plant • Gate entry verification • Material unloaded	• Live location tracking • Gate entry logged • Time in/out recorded • Quality check documented
5. Closure	Vendor BSP POC	• Final weighing at BSP • Document finalization • Vendor creates invoice • Payment processing	• Calculates net weight • Generates material receipt • Processes invoice • Completes request

■ USER ROLES & RESPONSIBILITIES

Role	Responsibilities	Key Rights/Permissions	Screens
1. Vendor (Industrial Partner)	• Create collection requests • Track vehicle status • Acknowledge loading • Manage invoices	• Full access to own requests • View vehicle location • Create/edit invoices • Upload documents	8 screens
2. Transporter	• Accept/reject requests • Assign vehicles & drivers • Fill gate entry forms • Manage fleet	• View all incoming requests • Assign resources • Update vehicle status • Access GPS tracking	7 screens

3. BSP POC (Operations)	• Approve/reject requests • Monitor operations • Generate reports • Day-to-day planning	• Approve all requests • View all collections • Access analytics • Reschedule pickups	5 screens
4. Gate Clerk	• Verify documents • Record gate entries • Link to purchase orders • Track in/out times	• Create gate entries • Verify vehicle documents • Assign entry numbers • View active vehicles	2 screens
5. Weighbridge Operator	• Record vehicle weights • Capture weight slip photos • Calculate net weights • Maintain records	• Enter weight data • Upload images • View weighing history • Generate weight reports	2 screens
6. System Admin	• Manage user accounts • Configure system settings • Monitor performance • Generate analytics	• Full system access • User management • System configuration • Audit logs access	4 screens

Total Application Screens: 28+

• Vendor Screens: 8 (Dashboard, Create Request, Track Request, Invoices, Profile, etc.)
• Transporter Screens: 7 (Dashboard, Request Management, Vehicle/Driver Management, GPS Tracking, etc.)
• BSP POC Screens: 5 (Dashboard, Approvals, Monitoring, Reports, Planning)
• Gate Clerk Screens: 2 (Gate Entry, Active Vehicles)
• Weighbridge Operator Screens: 2 (Weighing Interface, History)
• Admin Screens: 4 (User Management, System Config, Analytics, Logs)

Industrial Self-Collection App - Technical Overview

■ KEY FEATURES & CAPABILITIES

Feature Category	Description	Business Value
Mobile-First Design	Native Android & iOS apps for vendors and transporters with intuitive, user-friendly interfaces	Easy adoption, minimal training required
Real-Time GPS Tracking	Live vehicle location monitoring with route history, distance tracking, and ETA calculations	Complete transparency, better planning
Automated Workflows	Request creation, approvals, notifications, and status updates happen automatically	80% reduction in coordination time
Digital Weighbridge	Electronic weight recording at vendor site and BSP plant with photo documentation	Eliminates disputes, ensures accuracy
Invoice Management	Automated invoice generation using actual weight data with approval workflows	Faster payments, reduced errors
Multi-Channel Alerts	Push notifications, SMS, and email alerts for all stakeholders at each stage	No missed communications, real-time updates
Document Management	Digital storage of fitness certificates, gate entries, receipts, and compliance documents	Complete audit trail, easy compliance
Analytics Dashboard	Real-time metrics, collection statistics, vendor/transporter performance insights	Data-driven decisions, identify bottlenecks

■ EXPECTED BENEFITS & ROI

Metric	Current State	With App	Improvement
Coordination Time	2-4 hours per request	<30 minutes	80% reduction
Vehicle Utilization	60-65%	85-90%	+30% efficiency
Real-Time Visibility	None (phone calls)	100% GPS tracked	Complete transparency
Manual Errors	15-20% of requests	<2% of requests	90% reduction
Documentation	Paper-based	Digital with audit trail	100% compliant
Vendor Satisfaction	Moderate	High	Significant improvement

■ TECHNOLOGY & IMPLEMENTATION

Technology Stack	Implementation Timeline
• Mobile: React Native / Flutter • Backend: Node.js / Python • Database: PostgreSQL with 17 tables • GPS: Real-time tracking integration • Cloud: AWS / Google Cloud • Security: SSL, JWT authentication	• Phase 1 (Months 1-2): Core setup & APIs • Phase 2 (Months 3-4): Mobile apps & workflows • Phase 3 (Months 5-6): GPS, weighbridge, invoicing • Phase 4 (Month 7): Testing & pilot deployment • Phase 5 (Month 8+): Full rollout & support

Next Steps

1. Review and approve this proposal
2. Finalize specific customization requirements
3. Conduct stakeholder workshops for detailed screen designs
4. Begin development with Phase 1 (database & core APIs)
5. Pilot with 2-3 selected vendors before full rollout

This proposal outlines the complete Industrial Self-Collection App with detailed process flows, user roles, responsibilities, and screen counts. For technical documentation including database schema and ERD diagrams, please refer to the comprehensive documentation package.